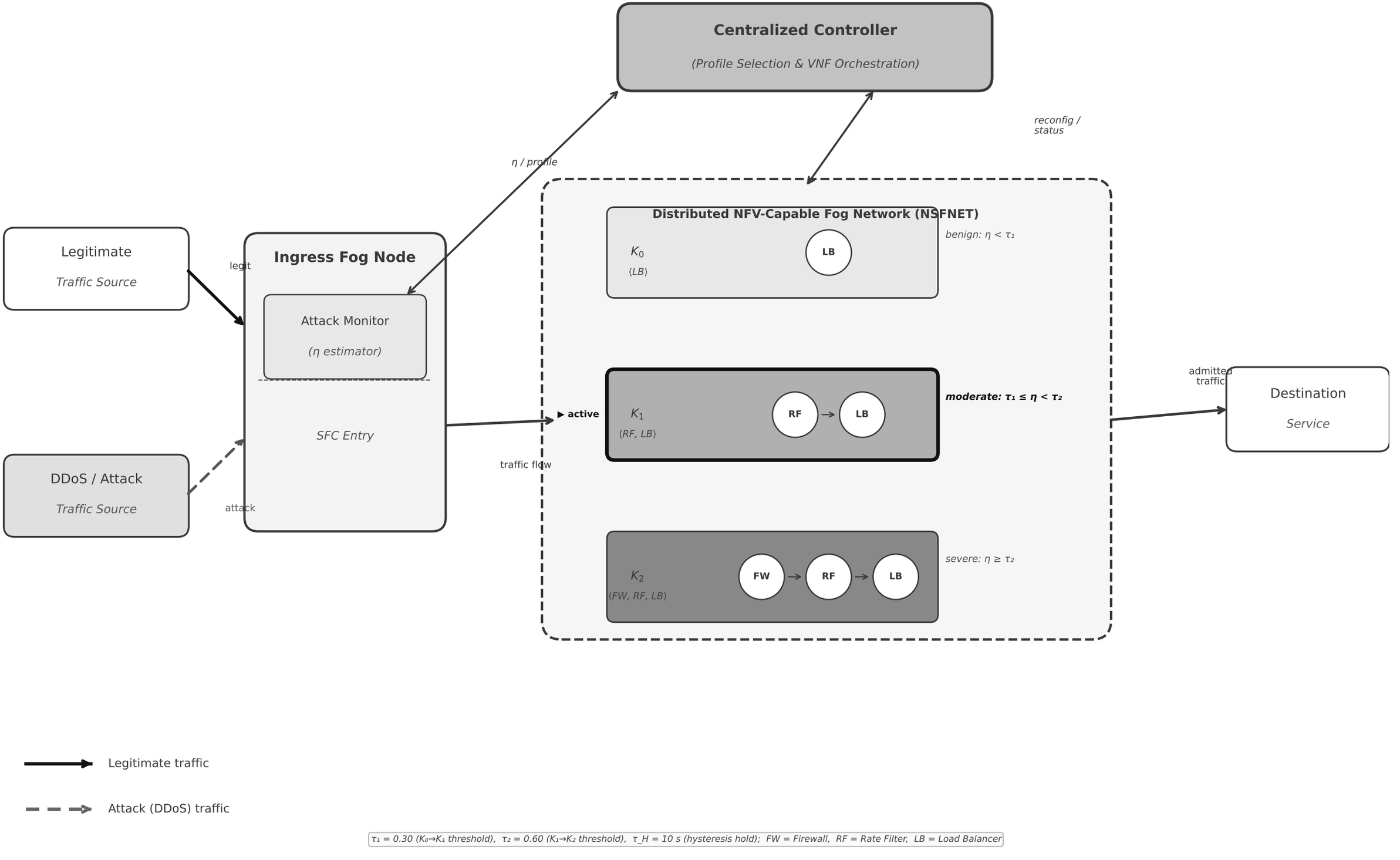
Centralized Controller
(Profile Selection & VNF Orchestration)
reconfig / status
η / profile
Legitimate
Traffic Source
legit
Ingress Fog Node
Attack Monitor
(η estimator)
SFC Entry
DDoS / Attack
Traffic Source
attack
traffic flow
Distributed NFV-Capable Fog Network (NSFNET)
K0
⟨LB⟩
LB
benign: η < τ1
active
K1
⟨RF, LB⟩
RF
LB
moderate: τ1 ≤ η < τ2
K2
⟨FW, RF, LB⟩
FW
RF
LB
severe: η ≥ τ2
admitted traffic
Destination
Service
Legitimate traffic
Attack (DDoS) traffic
τ1 = 0.30 (K0→K1 threshold), τ2 = 0.60 (K1→K2 threshold), τ_H = 10 s (hysteresis hold); FW = Firewall, RF = Rate Filter, LB = Load Balancer